1. Common General Diseases (High Frequency, Low Risk)

Malaria:
Malaria is a mosquito-borne infectious disease caused by *Plasmodium* parasites. Common symptoms include intermittent high fever with chills and sweating, headache, nausea, vomiting, muscle pain, and fatigue. In some cases, anemia and splenomegaly may be observed due to red blood cell destruction.
Basic Symptom-to-Condition Mapping:
High fever with chills and sweating, along with headache and fatigue, may indicate a febrile parasitic infection such as malaria, particularly in endemic regions.

Viral Infection (General):
Viral infections are caused by various viruses and commonly affect the respiratory or gastrointestinal systems. Typical symptoms include fever, body aches, fatigue, headache, sore throat, cough, and nasal congestion. Most viral infections are self-limiting and resolve with supportive care.
Basic Symptom-to-Condition Mapping:
Fever accompanied by body aches, fatigue, and upper respiratory symptoms may suggest a general viral infection.

Common Cold:
The common cold is a mild viral respiratory infection. Symptoms generally include sneezing, runny or blocked nose, sore throat, mild cough, headache, and low-grade fever. It is usually short-lived and non-severe.
Basic Symptom-to-Condition Mapping:
Sneezing, nasal congestion, and sore throat with mild or no fever are commonly associated with a common cold.

Influenza (Flu):
Influenza is an acute viral respiratory illness characterized by sudden onset of high fever, chills, muscle pain, headache, fatigue, dry cough, and sore throat. Symptoms are generally more severe than the common cold.
Basic Symptom-to-Condition Mapping:
Sudden high fever combined with muscle pain, fatigue, and dry cough may indicate influenza rather than a mild respiratory infection.

Diabetes Mellitus:
Diabetes Mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels. Common symptoms include frequent urination, excessive thirst, increased hunger, unexplained weight loss, fatigue, and delayed wound healing.
Basic Symptom-to-Condition Mapping:
Frequent urination and excessive thirst, along with fatigue or unexplained weight loss, may be associated with abnormal glucose regulation.

Hypertension:
Hypertension is a condition marked by persistently elevated blood pressure. It is often asymptomatic but may present with headache, dizziness, fatigue, or blurred vision in advanced cases.
Basic Symptom-to-Condition Mapping:
Persistent headaches or dizziness without clear cause may be associated with elevated blood pressure, especially in at-risk individuals.

Anemia:
Anemia is a condition in which the blood has a reduced capacity to carry oxygen due to low hemoglobin levels. Symptoms include fatigue, weakness, pale skin, shortness of breath, and dizziness.
Basic Symptom-to-Condition Mapping:
Fatigue, pallor, and shortness of breath on exertion may be linked to reduced hemoglobin levels.

Asthma:
Asthma is a chronic respiratory condition involving airway inflammation and constriction. Common symptoms include wheezing, shortness of breath, chest tightness, and coughing, particularly at night or early morning.
Basic Symptom-to-Condition Mapping:
Recurrent wheezing and breathlessness, especially during night-time or physical activity, may indicate an underlying airway condition.

Thyroid Disorders:
Thyroid disorders involve abnormal thyroid hormone levels. Hypothyroidism may cause fatigue,

weight gain, cold intolerance, and dry skin, while hyperthyroidism may present with weight loss, heat intolerance, palpitations, and anxiety.

Basic Symptom-to-Condition Mapping:

Unexplained weight changes combined with temperature sensitivity or fatigue may be associated with thyroid hormone imbalance.

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2. Lifestyle and Chronic Conditions

Obesity:

Obesity is a condition characterized by excessive body fat accumulation that increases health risks. Common indicators include increased body mass index (BMI), reduced physical endurance, joint pain, and fatigue. Obesity is strongly associated with lifestyle factors such as poor diet and physical inactivity.

Basic Symptom-to-Condition Mapping:

Reduced stamina, joint discomfort, and elevated BMI may indicate excess body weight–related health risks.

Pre-Diabetes:

Pre-diabetes is a metabolic condition where blood glucose levels are higher than normal but not yet in the diabetic range. Symptoms are often absent, though mild fatigue, increased thirst, or frequent urination may occur.

Basic Symptom-to-Condition Mapping:

Mild fatigue combined with increased thirst or urination may suggest early glucose metabolism imbalance.

Dyslipidemia (Cholesterol Disorders):

Dyslipidemia refers to abnormal levels of lipids in the blood, including elevated cholesterol or triglycerides. It is often asymptomatic but increases cardiovascular risk.

Basic Symptom-to-Condition Mapping:

Absence of symptoms with abnormal lipid values on blood tests may indicate a lipid metabolism disorder.

Cardiovascular Risk Conditions:

Cardiovascular risk conditions include early-stage disorders that increase the likelihood of heart disease. Symptoms may include exertional breathlessness, mild chest discomfort, palpitations,

or fatigue.

Basic Symptom-to-Condition Mapping:

Shortness of breath or chest discomfort during physical activity may be associated with increased cardiovascular risk.

Fatty Liver Disease (Early Stage):

Fatty liver disease is characterized by excessive fat accumulation in liver cells, commonly linked to metabolic disorders. Early stages are usually asymptomatic, though fatigue or mild abdominal discomfort may occur.

Basic Symptom-to-Condition Mapping:

Unexplained fatigue or mild right-sided abdominal discomfort, combined with metabolic risk factors, may be associated with early fatty liver changes.

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3. Standard Diagnostic Tests and Normal Ranges

Diagnostic tests are commonly used to assess overall health status and to identify deviations from normal physiological functioning. These tests support clinical evaluation by providing measurable indicators rather than definitive diagnoses.

Complete Blood Count (CBC):

A CBC evaluates components of blood such as hemoglobin, red blood cells, white blood cells, and platelets. It is commonly used to assess conditions like anemia, infection, and overall immune status. Abnormal values may indicate inflammation, nutritional deficiencies, or blood-related disorders.

Blood Glucose Tests:

Blood glucose testing includes fasting blood sugar (FBS), postprandial blood sugar (PPBS), and HbA1c. These tests help assess how the body regulates sugar over time. Elevated values may indicate impaired glucose metabolism.

Lipid Profile:

A lipid profile measures cholesterol and triglyceride levels, including LDL, HDL, and total cholesterol. It is used to assess cardiovascular risk. Abnormal lipid levels may increase the likelihood of heart-related conditions.

Thyroid Function Tests:
Thyroid tests measure hormones such as TSH, T3, and T4 to evaluate thyroid gland activity. Deviations from normal ranges may reflect underactive or overactive thyroid function.

Liver Function Tests (LFT):
LFTs measure enzymes and proteins such as ALT, AST, bilirubin, and albumin. These tests help assess liver health and metabolic function.

Kidney Function Tests (KFT):
KFTs include measurements such as serum creatinine and blood urea nitrogen (BUN). These values are used to evaluate kidney filtration and waste removal efficiency.

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4. Preventive and Primary Healthcare Information

Preventive healthcare focuses on reducing the risk of disease through early identification, healthy lifestyle practices, and routine monitoring. This information supports informed decision-making and health awareness.

Routine Health Checkups:
Regular medical checkups help detect early abnormalities before symptoms become severe. Periodic screening is especially important for individuals with chronic conditions or family history of disease.

Vaccination Awareness:
Vaccinations help protect against preventable infectious diseases by strengthening immune response. Adherence to recommended vaccination schedules is an important public health measure.

Healthy Lifestyle Practices:
Balanced nutrition, regular physical activity, adequate sleep, and stress management contribute significantly to long-term health. These practices help reduce the risk of lifestyle-related diseases.

Early Medical Consultation:
Seeking timely medical advice when symptoms persist or worsen can prevent complications. Early consultation is particularly important for symptoms such as prolonged fever, unexplained weight loss, or persistent fatigue.

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5. Medical Report and Document Understanding

Medical reports and health documents contain structured information that may be difficult for patients to interpret. Basic understanding of these documents improves patient awareness and engagement.

Types of Medical Reports:
Common reports include blood test reports, imaging reports (X-ray, CT scan), prescriptions, discharge summaries, and diagnostic summaries. Each serves a different purpose in healthcare management.

Key Information Extraction:
Medical reports typically include patient details, test dates, measured values, reference ranges, and remarks. Values outside reference ranges are usually highlighted for attention.

Simple Report Summarization:
Summarization involves presenting report findings in clear, non-technical language. This may include identifying whether values fall within or outside normal limits without interpreting them as diagnoses.

Limitations of Automated Understanding:
Automated systems can assist in organizing and explaining report content but do not replace professional medical interpretation. Clinical decisions should always be made by qualified healthcare providers.

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6. Key Diagnostic Components and Reference Ranges

Medical reports often include numerical values that are interpreted by comparing them with standard reference ranges. The following components are commonly found in routine diagnostic reports and help in identifying potential abnormalities.

6.1 Complete Blood Count (CBC)

Hemoglobin (Hb):
Normal Range:
• Adult Male: 13.0–17.0 g/dL
• Adult Female: 12.0–15.0 g/dL
Indicates the oxygen-carrying capacity of blood.

Red Blood Cell Count (RBC):
Normal Range:
• Male: 4.5–5.9 million cells/µL
• Female: 4.1–5.1 million cells/µL
Reflects the number of red blood cells in circulation.

White Blood Cell Count (WBC):
Normal Range: 4,000–11,000 cells/µL
Associated with immune response and infection indicators.

Platelet Count:
Normal Range: 150,000–450,000 cells/µL
Plays a role in blood clotting.

6.2 Blood Glucose Parameters

Fasting Blood Sugar (FBS):
Normal Range: 70–99 mg/dL
Measured after overnight fasting.

Postprandial Blood Sugar (PPBS):
Normal Range: <140 mg/dL
Measured two hours after meals.

HbA1c:
Normal Range: <5.7%
Indicates average blood glucose levels over 2–3 months.

6.3 Lipid Profile Components

Total Cholesterol:
Normal Range: <200 mg/dL

Low-Density Lipoprotein (LDL):
Normal Range: <100 mg/dL

High-Density Lipoprotein (HDL):
Normal Range:
- Male: >40 mg/dL
- Female: >50 mg/dL

Triglycerides:
Normal Range: <150 mg/dL

6.4 Thyroid Function Tests

Thyroid Stimulating Hormone (TSH):
Normal Range: 0.4–4.0 mIU/L

Triiodothyronine (T3):
Normal Range: 80–200 ng/dL

Thyroxine (T4):
Normal Range: 5.0–12.0 µg/dL

6.5 Liver Function Tests (LFT)

Alanine Aminotransferase (ALT):
Normal Range: 7–56 U/L

Aspartate Aminotransferase (AST):
Normal Range: 10–40 U/L

Total Bilirubin:
Normal Range: 0.1–1.2 mg/dL

Albumin:
Normal Range: 3.5–5.0 g/dL

6.6 Kidney Function Tests (KFT)

Serum Creatinine:
Normal Range:
• Male: 0.7–1.3 mg/dL
• Female: 0.6–1.1 mg/dL

Blood Urea Nitrogen (BUN):
Normal Range: 7–20 mg/dL